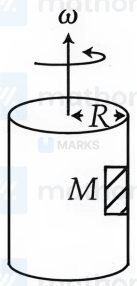
Q1. JEE Main 2026 (21 January Shift 2)

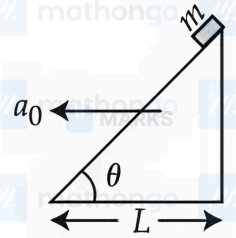
A large drum having radius R is spinning around its axis with angular velocity ω, as shown in figure. The minimum value of ω so that a body of mass M remains stuck to the inner wall of the drum, taking the coefficient of friction between the drum surface and mass M as μ, is :



(1) $\sqrt{\frac{\mu g}{R}}$ (2) $\sqrt{\frac{g}{2\mu R}}$ (3) $\sqrt{\frac{2g}{\mu R}}$ (4) $\sqrt{\frac{g}{\mu R}}$

Q2. JEE Main 2026 (28 January Shift 2)

A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L. Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is ____.



(1) $\sqrt{\frac{2L}{g\sin\theta - a_0\cos\theta}}$ (2) $\sqrt{\frac{4L}{g\cos^2\theta - a_0\sin\theta\cos\theta}}$

(3) $\sqrt{\frac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$ (4) $\sqrt{\frac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

Q3. JEE Main 2026 (24 January Shift 2)

A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30°. Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is ____.

(1) $\frac{\sqrt{3}}{2}$mg (2) mg (3) $\frac{1}{2}m$ g (4) $\sqrt{3}$mg

Q4. JEE Main 2026 (23 January Shift 2)

A block is sliding down on an inclined plane of slope θ and at an instant $t = 0$ this block is given an upward momentum so that it starts moving up on the inclined surface with velocity u. The distance (S) travelled by the block before its velocity become zero, is ____.

(g = gravitational acceleration)

(1) $\frac{2u^2}{\cos\theta}$ (2) $\frac{u^2}{2g\sin\theta}$ (3) $\frac{u^2}{2g\cos\theta}$ (4) $\frac{u^2}{\sqrt{2}g\cos\theta}$

Q5. JEE Main 2026 (21 January Shift 1)

A 4 kg mass moves under the influence of a force $\vec{F} = \left(4t^3\hat{i} - 3t\hat{j}\right)$N where t is the time in second. If mass starts from origin at $t = 0$, the velocity and position after $t = 2$ s will be:

(1) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} - \hat{j}$

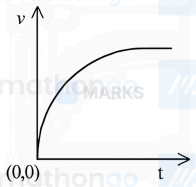
(2) $\vec{v} = 3\hat{i} + \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} + \hat{j}$

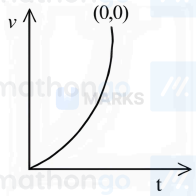
(3) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} - \hat{j}$

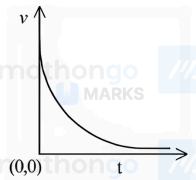
(4) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$

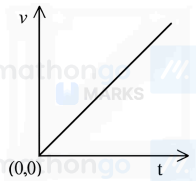
Q6. JEE Main 2026 (28 January Shift 1)

A particle of mass m falls from rest through a resistive medium having resistive force, $F = -kv$, where v is the velocity of the particle and k is a constant. Which of the following graphs represents velocity (v) versus time (t)?

(1)


(2)


(3)


(4)


Q7. JEE Main 2026 (28 January Shift 1)

A block of mass 5 kg is moving on an inclined plane which makes an angle of 30° with the horizontal. Friction coefficient between the block and inclined plane surface is $\frac{\sqrt{3}}{2}$. The force to be applied on the block so that the block will move down without acceleration is ____ N.

$\left(g = 10\ \mathrm{m/s^2}\right)$.

(1) 25 (2) 7.5 (3) 12.5 (4) 15

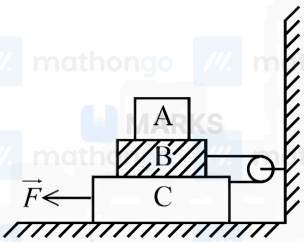
Q8. JEE Main 2026 (24 January Shift 1)

A spring of force constant 15 $\mathrm{N/m}$ is cut into two pieces. If the ratio of their length is $1:3$, then the force constant of smaller piece is ____ $\mathrm{N/m}$.

(1) 60 (2) 15 (3) 20 (4) 45

Q9. JEE Main 2026 (24 January Shift 1)

In the given figure the blocks A, B and C weigh 4 kg, 6 kg and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5. The force $\vec{F}$ required to slide the block C with constant speed is ____ N.
(Use $g = 10\ \mathrm{m/s^2}$)



ANSWERS AND SOLUTIONS

1. (4) **2.** (4) **3.** (1) **4.** (2) **5.** (3) **6.** (1) **7.** (3) **8.** (1)

9. 210

1. **(4)** Consider the mass M in the frame of the rotating drum. The forces acting on the mass are:

1. Weight Mg acting vertically downwards.
2. Normal reaction N from the wall acting towards the center, which provides the centripetal force: $N = M\omega^2 R$.
3. Frictional force f acting vertically upwards to balance the weight.

For the mass to remain stuck to the wall, the frictional force must balance the weight: $f = Mg$.

The maximum available static friction is $f_{max} = \mu N = \mu M\omega^2 R$.

To prevent slipping, we must have $f_{max} \geq Mg$.

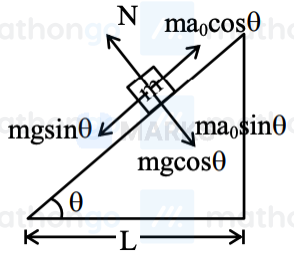
$\mu M\omega^2 R \geq Mg$

$\omega^2 \geq \frac{g}{\mu R}$

$\omega \geq \sqrt{\frac{g}{\mu R}}$

The minimum value of ω is $\sqrt{\frac{g}{\mu R}}$.

2.



(4)

$mg\sin\theta - ma_0\cos\theta = ma$

$a = g\sin\theta - a_0\cos\theta$

Now using,

$$S = ut + \frac{1}{2}a_{down}t^2$$

$$\frac{L}{\cos\theta} = \frac{1}{2}(g\sin\theta - a_0\cos\theta)t^2$$

$$t = \sqrt{\frac{2L}{g\sin\theta\cos\theta - a_0\cos^2\theta}}$$

$$t = \sqrt{\frac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$$

3. **(1)** A chain of mass m hangs between two fixed points at the same level with inclination 30° at support points.

Consider equilibrium of one half of the chain (mass $m/2$):

Horizontal equilibrium: $T_0 = T_1\cos(30^\circ) = T_1 \cdot \frac{\sqrt{3}}{2}$

Vertical equilibrium: $T_1\sin(30^\circ) = \frac{m}{2}g$

From vertical equation: $T_1 \cdot \frac{1}{2} = \frac{mg}{2}$, giving $T_1 = mg$

Substituting into horizontal equation: $T_0 = mg \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}mg$

4. **(2)** $a = -g\sin\theta$

$V^2 = U^2 + 2as$

$0 = u^2 - 2g\sin\theta \cdot s$

$s = \dfrac{u^2}{2g\sin\theta}$

5. **(3)** Given $\vec{F} = (4t^3\hat{i} - 3t\hat{j})$ N, m = 4 kg

$\vec{a} = \frac{\vec{F}}{m} = t^3\hat{i} - \frac{3t}{4}\hat{j}$

Integrating with $\vec{v}(0) = 0$:

$\vec{v} = \frac{t^4}{4}\hat{i} - \frac{3t^2}{8}\hat{j}$

At $t = 2$: $\vec{v} = \frac{16}{4}\hat{i} - \frac{12}{8}\hat{j} = 4\hat{i} - \frac{3}{2}\hat{j}$

Integrating with $\vec{r}(0) = 0$:

$\vec{r} = \frac{t^5}{20}\hat{i} - \frac{t^3}{8}\hat{j}$

At $t = 2$: $\vec{r} = \frac{32}{20}\hat{i} - \frac{8}{8}\hat{j} = \frac{8}{5}\hat{i} - \hat{j}$

6. **(1)** The particle experiences downward gravitational force (mg) and upward resistive force (-kv).

The equation of motion is $m\frac{dv}{dt} = mg - kv$.

Initially at $t = 0$, $v = 0$ gives maximum acceleration $a_0 = g$.

As velocity increases, the resistive force increases, reducing the net acceleration.

Eventually the particle reaches terminal velocity $v_t = \frac{mg}{k}$ when acceleration becomes zero.

The velocity-time graph shows an increasing curve starting from origin with decreasing slope, asymptotically approaching the terminal velocity.

This behavior matches option (1), which displays a curve starting at (0,0) with steep initial slope that gradually flattens.

7. **(3)** For motion without acceleration, net force = 0.

Component of weight along incline: $mg\sin(30^\circ) = 5 \times 10 \times 0.5 = 25$ N (down).

Normal force: $N = mg\cos(30^\circ) = 50 \times \frac{\sqrt{3}}{2} = 25\sqrt{3}$ N.

Friction force opposing downward motion:

$f = \mu N = \frac{\sqrt{3}}{2} \times 25\sqrt{3} = 37.5$ N (up).

For equilibrium along the incline, applying force F down the incline:

$mg\sin\theta + F = f$, so $25 + F = 37.5$, giving $F = 12.5$ N.

8. **(1)** Spring force constant is inversely proportional to length: $k \propto \frac{1}{L}$

So $kL =$ constant for a given spring material and structure.

Original spring: $k = 15$ N/m, length $= L$.

Smaller piece has length $\frac{L}{4}$ (since length ratio is 1:3, smaller is $\frac{1}{4}$ of total).

For smaller piece: $k_1 \times \frac{L}{4} = 15 \times L$

$k_1 = 15 \times 4 = 60$ N/m

9. **(210)** For the 8 kg block (C) to move with constant velocity, $F_{net} = 0$.

Friction forces acting:

- Between C and ground: $f = \mu(m_A + m_B + m_C)g = 0.5 \times (4 + 6 + 8) \times 10 = 90$ N
- Between C and B: $f = \mu(m_A + m_B)g = 0.5 \times (4 + 6) \times 10 = 50$ N

For the 6 kg block (B):

$T = f_{\text{B on ground}} + f_{\text{A on B}}$

$T = \mu(m_A + m_B)g \cdot \frac{m_B}{m_A + m_B} + \mu \cdot m_A \cdot g$

$T = 20 + 50 = 70$ N

For the 8 kg block (C):

$F = 90 + T + 50 = 90 + 70 + 50 = 210$ N